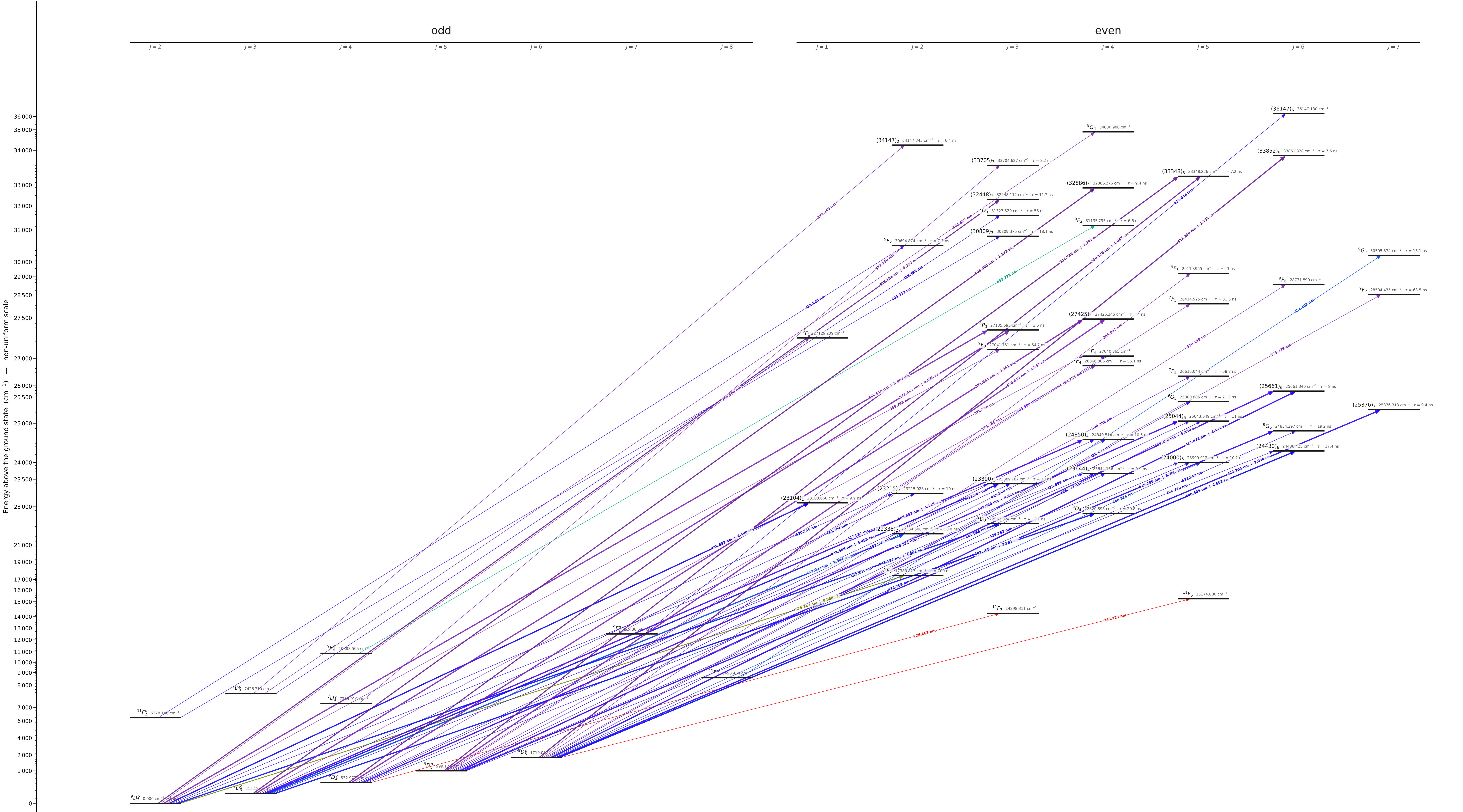
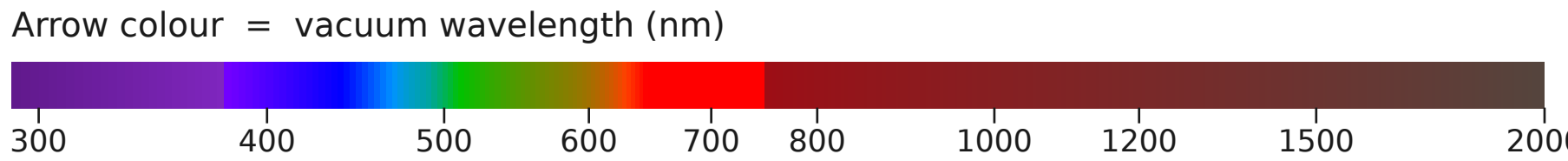


Gadolinium

Gd (Gd I). Strongest classified transitions below 2 μm (NIST): 53 levels, 60 lines.

Gd+ ionisation limit 49601.45 cm⁻¹ ≈ 6.14980 eV (201.607 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U|er|U'\rangle$ in ea.
Matrix elements are measured or NIST values; * = high-accuracy theory, = = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n ≈ 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[U er U'] (ea)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁹ D ₅ - ⁹ D ₃	368.5183	368.4134	813.5077	3.967	9.100e+07	45.47	31.9	proposed laser-cooling line
⁹ D ₅ - ⁹ D ₁	371.4631	371.3575	807.0584	4.036	9.200e+07	45.47	32.2	371 nm repumper for the proposed 368 nm cooling scheme
⁹ D ₅ - (25376) ₁	422.7038	422.5848	709.2258	7.054	8.900e+07	16.93	83.7	proposed laser-cooling line
⁹ D ₅ - ⁹ D ₃	443.1873	443.0629	676.4464	3.004	3.000e+07	11.62	41.1	443.06 nm
⁹ D ₅ - (22335) ₂	452.0922	451.9654	663.1224	2.944	3.800e+07	14.74	41	451.96 nm
⁹ D ₅ - ⁹ F ₂	575.3466	575.1871	521.0641	0.568	6.855e+05	0.2274	48	first step
⁹ D ₅ - ¹¹ F ₃	726.4626	726.2624	412.6743	-	-	-	-	726.266 nm spin-forbidden line 9D ⁴ - 11F3 studied as a c...
⁹ D ₅ - ¹¹ F ₅	743.2229	743.0183	403.3681	-	-	-	-	743.014 nm spin-forbidden line 9D ⁶ - 11F5 studied as a c...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁵⁷Gd hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
⁹ D ₂	47.959(8)	191.16(7)	F=1/2: -48.55 F=3/2: -143.88 F=5/2: -143.46 F=7/2: +191.67
⁹ D ₃	6.4456(22)	-433.234(13)	F=3/2: -298.61 F=5/2: +42.42 F=7/2: +216.62 F=9/2: -79.30
⁹ D ₄	-8.997(5)	-375.88(4)	F=5/2: -117.11 F=7/2: +109.82 F=9/2: +159.95 F=11/2: -147.95
⁹ D ₅	-15.095(4)	44.74(4)	F=7/2: +155.25 F=9/2: +58.23 F=11/2: -32.99 F=13/2: -102.03
⁹ D ₆	-16.057(6)	623.27(6)	F=9/2: +416.50 F=11/2: -61.37 F=13/2: -257.81 F=15/2: +11.30
¹¹ F ₂	-227.1080(20)	-	F=1/2: +1021.99 F=3/2: +681.32 F=5/2: +113.55 F=7/2: -681.32
¹¹ F ₃	-106.1240(20)	-	F=13/2: +1432.67 F=15/2: +636.74 F=17/2: -265.31 F=19/2: -1273.49
¹¹ F ₄	-77(5)	590(4)	F=3/2: +813.30 F=5/2: +179.68 F=7/2: -294.75 F=9/2: -197.32
¹¹ F ₅	84(4)	967(4)	F=7/2: -335.94 F=9/2: -587.13 F=11/2: -303.02 F=13/2: +871.08
⁹ F ₂	95.100	347.000	F=1/2: -124.33 F=3/2: -285.30 F=5/2: -264.43 F=7/2: +372.05

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
⁹ D ₂ - ⁷ F ₅	364.753	160-156	-2398(90)
⁷ D ₃ - ⁹ G ₄	364.827	160-156	-2998(90)
⁹ D ₂ - ⁹ F ₅	364.952	160-156	-3328(90)
⁹ D ₃ - ⁹ F ₃	368.606	160-156	-3628(90)
⁹ D ₂ - ⁹ F ₃	369.798	160-156	-3598(90)
⁹ D ₂ - ⁹ F ₆	370.199	160-156	-4077(90)
⁹ D ₃ - ⁹ F ₄	372.776	160-156	-3298(90)
⁹ D ₂ - ⁹ F ₇	373.338	160-156	-2998(90)
⁷ D ₃ - (34147) ₂	374.243	160-156	-2998(90)
⁷ D ₄ - (33705) ₃	377.790	160-156	-3328(90)
⁹ D ₂ - ⁷ F ₄	379.746	160-156	-3418(90)
⁹ D ₃ - ⁹ F ₄	383.099	160-156	-3687(90)
⁹ D ₂ - ⁷ F ₃	390.382	160-156	-3927(90)
⁹ D ₃ - (24850) ₄	405.937	160-156	-3298(90)
⁹ D ₁ - (25044) ₅	407.986	160-156	-3658(90)
¹¹ F ₃ - (30809) ₃	409.312	160-156	-2638(90)
¹¹ F ₂ - ⁹ F ₂	411.240	160-156	-1199(150)
⁹ D ₃ - (24850) ₄	411.243	160-156	-3328(90)
⁹ D ₃ - (25044) ₅	415.895	160-156	-3747(90)
⁷ D ₃ - ⁷ D ₃	418.396	160-156	-3897(90)
⁹ D ₃ - ⁹ G ₆	419.196	160-156	-4227(90)
⁹ D ₃ - (24850) ₄	419.280	160-156	-2998(90)

Shift = ν(first isotope) - ν(second isotope).